Maneesh Punetha
Assistant Professor, School of Mechanical Sciences
Indian Institute of Technology Bhubaneswar
Odisha, India – 752050

+91-8960544696 ‖ maneeshp@iitbbs.ac.in ‖ maneesh40@gmail.com
Webpage: https://maneeshpunetha.github.io/index.html
Google Scholar: https://scholar.google.co.in/citations?user=8GQ6AHwAAAAJ&hl=en
Research Gate: https://www.researchgate.net/profile/Maneesh_Punetha

Key Research interests

Nuclear Thermal Hydraulics, Nuclear Power Safety, Hydrogen Storage and Safety, and Phase Change Heat Transfer.

Academic Qualifications

- **Ph. D.** (2014-2020) – Mechanical Engineering – **Indian Institute of Technology Kanpur**, Uttar Pradesh, India

  CGPA: 8.67 and **Outstanding Ph.D. Thesis Award, 2020, IIT Kanpur**
  Thesis: *Containment Thermal Hydraulic Studies towards Understanding Post-Severe Nuclear Accident Scenarios*
  Supervisor: Prof. Sameer Khandekar

- **M. Tech.** (2012-2014) – Thermal System Design – **Sardar Vallabhbhai National Institute of Technology,** Gujarat, India

  CGPA: 9.65
  Thesis: *Analysis of Dispersion of Heated Effluent in Lake Scenario*
  Supervisor: Prof. Jyotirmay Banerjee

- **B. Tech.** (2007-2011) – Mechanical Engineering – **Government College of Engineering Amravati**, Maharashtra, India

  CGPA: 8.45
  Project: *Optimization of number of turns of Closed Loop Pulsating Heat Pipe (CLPHP)*
  Supervisor: Dr. P.R. Pachghare

Post- Ph. D. Research Experience

- **December 2023 – Present,** as Assistant Professor, School of Mechanical Sciences, IIT Bhubaneswar, Odisha, India

- **June 2021 - December 2023,** as Postdoctoral Fellow, Division of Nuclear Power Safety, School of Engineering Sciences, KTH Royal Institute of Technology, Sweden

  Supervisor: Prof. Sevostian Bechta

- **August 2020 - June 2021**, as Research Associate, Department of Mechanical Engineering, IIT Kanpur, Kanpur, India

  Supervisor: Prof. Sameer Khandekar

Industrial Experience

- **July 2011 - August 2012,** as Assistant Manager (Health, Safety and Environment) in Essar Steel India Limited, Hazira, Gujarat

  Attained six sigma yellow belt certification

Academic Achievements/Fellowships

- Received "**Outstanding Ph.D. Thesis Award - 2020**" of IIT Kanpur on 53rd Convocation.
- "Student Best Poster Award" for 'Effect of surface inclination on film condensation heat transfer in the presence of non-condensable gases' at 27th International Conference on Nuclear Engineering (ICONE27), Ibaraki, Japan, May 19-24, 2019.
- International travel grant from IIT Kanpur for attending the 27th International Conference on Nuclear Engineering (ICONE27), May 19-24, 2019 at Tsukuba International Congress Center, Tsukuba, Ibaraki, Japan.
- Received higher education scholarship from MHRD, Government of India to pursue Ph.D. at Indian Institute of Technology Kanpur, Kanpur (U. P.), India (2014-2019) and M.Tech. at Sardar Vallabhbhai National Institute of Technology, Surat (Gujarat), India (2012-2014).
- Awarded "Times of India Merit Scholarship" in M. Tech. for academic year of 2012-13.
- Awarded Merit scholarship by Army Welfare Education Society (AWES), New Delhi, India in all four years of B. Tech. from 2007-2011.

Teaching Experience

- **Teacher** (Indian Institute of Technology Bhubaneswar, Odisha, India)
  - ME6L162 Nuclear Power Generation and Safety
  - ME6L112 Two-Phase Flow Heat Transfer
  - ME3L006 Refrigeration and Air Conditioning
  - ME3L501 Applied Thermodynamics
  - ME2L501 Elements of Mechanical Engineering
  - ME6P001 Mechanical Systems Simulation Laboratory - I
  - ME1P001 / CE1P001 Introduction to Manufacturing Processes (Laboratory)
- **Teacher** (KTH Royal Institute of Technology, Stockholm, Sweden)
  - SH1015 HT21 Applied Modern Physics
  - FSH3001 HT21-1 Nuclear Power Safety Engineering Project
- **Teaching Assistant** (Indian Institute of Technology Kanpur, Kanpur, UP, India)
  - ME639 Liquid-Vapour Phase-Change Phenomena
  - ME304A Refrigeration and Air Conditioning

List of Publications

1. Pathy P. P., and **Punetha M.**, CFD-based Thermal Hydraulic Modelling of Small Modular Reactor Containment during a Postulated Steam Leakage Scenario, Nuclear Engineering and Design, Accepted for publication. **(Web of Science: Q1)**
2. **Punetha M.**, Yadav M. K., Jain S., Khandekar S., and Sharma P. K., Thermal-Hydraulic Test Facility for Nuclear Reactor Containment: Engineering Design Methodology and Benchmarking, Progress in Nuclear Energy, Vol. 138, pp. 1-20, August 2021. DOI: 10.1016/j.pnucene.2021.103837 **(Web of Science: Q1)**
3. **Punetha M.**, Yadav M. K., Khandekar S., Sharma P. K., and Ganju S., Intrinsic Transport and Combustion Issues of Steam-Air-Hydrogen Mixtures in Nuclear Containments, International Journal of Hydrogen Energy, Vol. 45 (4), pp. 3340-3371, 2020. DOI: 10.1016/j.ijhydene.2019.11.179 **(Web of Science: Q1)**
4. **Punetha M.**, Choudhary A., and Khandekar S., Stratification and Mixing Dynamics of Helium in an Air-Filled Confined Enclosure, International Journal of Hydrogen Energy, Vol. 43 (42), pp. 19792-19809, 2018. DOI: 10.1016/j.ijhydene.2018.08.168 **(Web of Science: Q1)**
5. **Punetha M.**, and Khandekar S., A CFD based Modeling Approach for Predicting Steam Condensation in the Presence of Non-condensable Gases, Nuclear Engineering and Design, Vol. 324, pp. 280-296, 2017. DOI: 10.1016/j.nucengdes.2017.09.007. **(Web of Science: Q1)**
6. Zhao L., **Punetha M.**, Ma W., Konovalenko A., and Bechta S., Simulation of Melt Spreading over Dry Substrates with the Moving Particle Semi-implicit Method, Nuclear Engineering and Design, Vol. 405, pp. 1-12, April 2023. **(Web of Science: Q1)**
7. Yadav M. K., **Punetha M.**, Bhanawat A., Khandekar S., and Sharma P. K., Steam Condensation Heat Transfer during Initial Blow-down Period of a Severe Nuclear Accident, ASME Journal of Nuclear Engineering and Radiation Science, Vol. 6 (4), pp. 1-9, October 2020. DOI: 10.1115/1.4046910 **(Web of Science: Q3)**
8. Bhanawat A., Yadav M. K., **Punetha M.**, Khandekar S., and Sharma P. K., Effect of Surface Inclination on Filmwise Condensation Heat Transfer During Flow of Steam–Air Mixtures, ASME Journal of Thermal Science and Engineering Applications, Vol. 12 (4), pp. 1-12, August 2020. DOI: 10.1115/1.4046867 **(Web of Science: Q1)**
9. Fang D., Deng Y., Xiang Y., **Punetha M.**, Zhao L., and Ma W., Experimental and numerical studies on spray cooling of a downward-facing surface under partial coverage of multi-nozzle sprays, Nuclear Engineering and Design, Vol. 428, pp. 1-16, November 2024. DOI: 10.1016/j.nucengdes.2024.113525 **(Web of Science: Q1)**
10. Zhao L., **Punetha M.**, Ma W., Bechta S., Isaksson P., Lomperski S. W., Farmer M. T., and Licht J. R., Application of moving particle semi-implicit method on simulating melt spreading within OECD/ROSAU project, Nuclear Engineering and Design, Vol. 427, pp. 1-11, October 2024. DOI: 10.1016/j.nucengdes.2024.113447 **(Web of Science: Q1)**

Book Chapters

1. Yadav M.K., **Punetha M.**, Bhanawat A., Khandekar S., and Muralidhar K., Measurement of Condensation Heat Transfer, in 'Drop Dynamics and Dropwise Condensation on Textured Surfaces', Mechanical Engineering Series, Springer, 2020. DOI: 2019-20 10.1007/978-3-030-48461-3_13
2. **Punetha M.**, Thermal Pollution: Mathematical Modelling and Analysis, in 'Environmental Contaminants', Energy, Environment and Sustainability, Springer, Singapore, 2018. DOI: 10.1007/978-981-10-7332-8_18

Peer Reviewed Conference Proceedings (published/presented)

1. Pathy P., and **Punetha M.**, Thermal Hydraulic Study for Passive Safety of Small Modular Reactor (SMR) Containment. 11th International and 51st National Conference on Fluid Mechanics and Fluid Power (FMFP- 2024), Aligarh Muslim University (AMU), Aligarh, Uttar Pradesh, 21st to 23rd December 2024.
2. **Punetha M.**, and Khandekar S., (Invited) Containment Thermal Hydraulics Study for a Nuclear Power Plant Disasters. 4th Biennial International Conference on Future Learning Aspects of Mechanical Engineering (FLAME - 2024), Amity University, Noida, Uttar Pradesh, 31st July to 2nd Aug 2024.
3. **Punetha M.**, Komlev, A., Konlvalenko, A., and Bechta, S., Thermo-hydrodynamic of Corium Simulant Melt Spreading: A small scale model study. '11th International Conference on Multiphase Flow (ICMF-2023)', Kobe, Japan. 2-7, April 2023.
4. **Punetha M.**, Kulkarni S., Yadav M.K., and Khandekar S., A CFD Study on Coupled Issues of Hydrogen Distribution and Steam Condensation Inside Thermal Hydraulic Test facility for Containment (THYCON), 25th National and 3rd International ISHMT-ASTFE Heat and Mass Transfer Conference, IIT Roorkee, Uttarakhand, India, December 28-31st, 2019.
5. **Punetha M.**, Yadav M.K., Bhanawat A., and Khandekar, S., Steam Condensation Heat Transfer inside Reactor Containment during the Initial Transient of a Severe Accident, Proceedings of 27th International Conference on Nuclear Engineering (ICONE27), Tsukuba, Ibaraki, Japan, May 18-24, 2019. DOI: 10.1299/jsmeicone.2019.27.2166
6. Bhanawat A., **Punetha M.**, Yadav M.K., and Khandekar, S., Effect of Surface Inclination on Film Condensation Heat Transfer in the Presence of Air, Proceedings of 27th International Conference on Nuclear Engineering, (ICONE27), Tsukuba, Ibaraki, Japan, May 18-24, 2019. DOI: 10.1299/jsmeicone.2019.27.2133
7. Kulkarni S., **Punetha M.**, Choudhary A., and Khandekar S., Effect of Stratification and Natural Circulation on Steam Condensation in Presence of Non-Condensable Gases, Proceedings of 5th International Conference on Computational Methods for Thermal Problems (ThermaComp - 2018), IISc Bangalore, Karnataka, India, pp. 480-483, July 9-11, 2018. ISSN: 23055995
8. **Punetha M.**, Choudhary A., Khandekar S. and Sharma P., Helium Stratification and Mixing Studies in a Fully Enclosed Chamber, 24th National Heat and Mass Transfer Conference and 2nd International ISHMT-ASTFE Heat and Mass Transfer Conference, BITS Hyderabad, Telangana, India, December 27-30, 2017.
9. **Punetha M.**, and Khandekar S., Study of Film-wise Condensation inside Closed Containment using Wall Condensation Model (WCM), 6th International and 43th National Conference on Fluid Mechanics and Fluid Power (FMFP2016), Motilal Nehru National Institute of Technology, Allahabad, Uttar Pradesh, India, December 15-17, 2016.
10. **Punetha M.**, Thaker, J. P., and Banerjee J., Experimental and Numerical Analysis of Dispersion of Heated Effluent from Power Plants, 5th International and 41th National Conference on Fluid Mechanics and Fluid Power (FMFP2014), Indian Institute of Technology Kanpur, Uttar Pradesh, India, December 12-17, 2014.
11. **Punetha M.**, Roopchandani C., and Banerjee J., Analysis for dispersion of thermal effluent from Thermonuclear Power Plant, 40th National Conference on Fluid Mechanics and Fluid Power (FMFP2013), National Institute of Technology Hamirpur, Himachal Pradesh, India, December 12-14, 2013.

Sponsored Projects at IIT Bhubaneswar:

- As **Principal Investigator**
  1. Thermal Hydraulic Study for Passive Safety of Small Modular Reactor (SMR) Containment (Dec 2024 - Ongoing)

Amount: Rs. 2.394 Million INR,
Sponsoring Agency: SRIC, IIT Bhubaneswar, Odisha, India

2. Gas stratification model development and Benchmarking of containment module of PRABHAVINI Code (Mar 2026 - Ongoing)

   Amount: Rs. 2.87 Million INR
   Sponsoring Agency: Board of Research in Nuclear Sciences (BRNS), DAE, India
   Sanction order no. 41/14/08/2025-BRNS

- As **Co Principal Investigator**

3. Funding of testing facilities, infrastructure, and institutional support for development of Standards and Regulatory framework under the National Green Hydrogen Mission (will commence from Aug 2026)

   Amount: Rs. 251 Million INR
   Sponsoring Agency: Ministry of New and Renewable Energy (MNRE), Govt. of India
   Sanction order no. 353/13/2024-New Technology.

Sponsored Projects at Other Institutions:

- At **KTH Royal Institute of Technology** (as Postdoctoral research)

1. Reduction of Severe Accident Uncertainties (ROSAU) for Reactor Core-Debris Cooling in Shallow Containment Cavity (Jun 2021 - Dec 2023)

   Sponsoring Agency: OECD Nuclear Energy Agency (NEA)

   Highlights: Developed CoSMUS (Corium Simulant Melt Underwater Spreading) — a new experimental facility with advanced measurement capabilities to study the hydrodynamic and thermal behavior of corium spreading on a water-pooled containment cavity floor, simulating Nordic PWR conditions; Investigated corium spreading phenomenology and support numerical code development for severe accident analysis.

2. In-Vessel Melt Retention (IVMR) for Reactor Core-Melt Cooling in Reactor vessel lower head (Jun 2021 - Dec 2023)
   Sponsoring Agency: Swedish Radiation Safety Authority (SSM).

   Highlights: The research is being conducted in two parts (a) Radiation Modelling of In-vessel Melt Coolability, (b) Spray cooling for In-Vessel Melt Retention (IVMR). The goal of this research is to increase confidence in the IVR strategy and guide modification in existing reactors and new designs.

- At **Indian Institute of Technology Kanpur** (as Postdoctoral and Ph.D. Research)

1. Studies on heat transfer during condensation of steam-hydrogen mixtures inside closed containment (Oct 2015 - Apr 2021)

   Sponsoring Agency: Bhabha Atomic Research Center (BARC), Mumbai, India.

   Highlights: Designed, fabricated, and commissioned THYCON (Thermal-Hydraulic test facility for Containment) — a large-scale single-compartment facility simulating Fukushima Daiichi 2011-type accidents in Indian BWRs; Installed and calibrated a 20-port online mass spectrometry system (Hiden Analytical®) for spatial composition analysis of multi-component gas mixtures (steam, air, and hydrogen); Implemented an inverse technique-based high heat flux measurement system for adverse operating conditions; Investigated the coupled behavior of steam condensation and hydrogen accumulation through experimental studies.

2. Local heat transfer coefficient during film condensation of steam hydrogen mixtures (Apr 2015 - Mar 2018)

   Sponsoring Agency: Board of Research in Nuclear Sciences (BRNS), Mumbai, India.

Highlights: Designed, fabricated, and comissioned CONFLO — an experimental facility for condensation studies in the presence of non-condensable gases; Conducted experiments to investigate steam condensation heat transfer in air and air-helium environments, spanning for a range of parameters relating various stages of severe nuclear accident progression.

PhD Thesis Supervision: 1 (completed) + 2 (ongoing)

1. July 2021 - Jun 2024 **Dr. Lu Zhao** Role: Co-supervisor

   Thesis: Simulation of Melt Infiltration and Spreading using Moving Particle Semi-implicit Method (jointly with Prof. Sevostian Bechta, Prof. Weimin Ma, and Dr. Alexander Konovalenko)

2. Aug 2021 - Ongoing **Mr. Md Monzur Hossain Khan** Role: Co-supervisor

   Thesis: Radiation Modelling of Reactor Pressure Vessel Lower Head (jointly with Prof. Sevostian Bechta, and Dr. Andrei Komlev)

3. Jul 2024 – Ongoing **Mr. Prabhu Prasad Pathy** Role: Sole Supervisor

   Thesis: Thermal hydraulics study of Small Modular Reactor (SMR) containment in the event of severe accident

M.Tech. Thesis Supervision: 3 (completed) + 4 (ongoing)

1. May 2024 – May 2025 **Mr. Latesh Govindrao Munde** Role: Sole Supervisor

   Thesis: Thermo-hydrodynamic study of corium in reactor cavity

2. May 2024 – May 2025 **Mr. Theeda Ravindra** Role: Sole Supervisor

   Thesis: Effect of radiation on reactor pressure vessel's molten pool

3. May 2024 – May 2025 **Mr. Kanad Mitra** Role: Sole Supervisor

   Thesis: Minimization of boiloff losses for liquid hydrogen storage

4. May 2025 – Ongoing **Mr. Pawar Rohit Naik** Role: Sole Supervisor

   Thesis: Enhancing condensation in presence of non-condensable gasses

5. May 2025 – Ongoing **Mr. Anurag Mani Tripathi** Role: Sole Supervisor

   Thesis: Stratification and mixing of gasses by lumped parameter approach

6. May 2025 – Ongoing **Mr. Kukkala Prasad** Role: Sole Supervisor

   Thesis: Cryogenic storage of liquid fuels

7. May 2025 – Ongoing **Mr. Anil Kumar Prajapati** Role: Sole Supervisor

   Thesis: Spray cooling for high heat flux application

Invited Speaker:

- Keynote lecture on 'Title: Underwater High-Temperature Melt Cooling Experiments Workshop on Digital Analysis of Flow Processes in Porous Materials for Geo-Energy Applications', Jan 15–16, 2026, IIT Bhubaneswar, India
- Lecture on 'Title: Small Modular Reactors (SMR): A future of nuclear energy' at Short Term Training Program titled 'Advances in Net Zero Energy Applications' under MMTTP, 07th November 2025, IIT (ISM) Dhanbad, JH, India.
- Keynote lecture on 'Title: Severe Accident Research in Nuclear Power Plant', at 3rd Biennial International Symposium on "Fluids and Thermal Engineering" (FLUTE 2025), 07th August 2025, Amity University, Noida (U.P), India

- Conducted technical session 'In the loving memory of Prof. Sameer Khandekar', FMFP-2024, 21st December 2024, Z.H. College of Engineering & Technology, Aligarh Muslim University, Aligarh (U.P), India.
- Lecture on 'Title: Rector Core Melt Relocation in a Severe Nuclear Accident' presented at Amity School of Engineering Technology, 20th December 2024, Amity University, Noida (U.P.), India.
- Lecture on 'Technology in Nuclear Energy and Space Systems' on National Science Day, 28th February 2024, Manas College of Science Technology and Management, Pithoragarh (Uttarakhand), India.
- Lecture on 'Title: Underwater Simulant-Corium Melt Spreading' at International Workshop on Fuel Coolant Interaction and Steam explosion, 3rd October 2023, IRSN Workshop on Fuel Coolant Interaction, Fontenay-aux-Roses (France).

Reviewer of Journals:

- Nuclear Engineering and Design (Elsevier)
- International Journal of Hydrogen Energy (Elsevier)
- International Communications in Heat and Mass Transfer (Elsevier)
- International Journal of Heat and Mass Transfer (Elsevier)
- ASME Journal of Thermal Science and Engineering Application (ASME)
- ASME Journal of Heat Transfer (ASME)
- Water Quality Research Journal (IWA Publishing)
- Sadhana- Academy Proceedings in Engineering Sciences (Springer)

Society Affiliations

- National Society of Fluid Mechanics and Fluid Power (NSFMFP)
- Indian Society for Heat and Mass Transfer (ISHMT)

Other Experience

- Website development of School of Mechanical Sciences, IIT Bhubaneswar, 2025-2026.
- Faculty advisor of B.Tech. 2024 Mechanical Engineering Batch of IIT Bhubaneswar, India from Jul 2024 – Ongoing.
- Co-PIC: Advance Thermal Engineering Laboratory, and Energy System laboratory at SMS, IIT Bhubaneswar, India (Mar 2024 – Mar 2026).
- Co-PIC: Heat Transfer Laboratory, and Thermofluid-II laboratory at SMS, IIT Bhubaneswar, India (Mar 2024 – Mar 2026).
- Institute Representative (IR) at JEE (Advanced) 2024 (25-26 May 2024)
- Maintained website of Mechanical Engineering Department, IIT Kanpur from Jul 2015-Dec 2019.

- (End of the document) -